

TALAS, Kyrgyzstan

WMO: 383450

Lat: 42.517N

Lon: 72.217E

Elev: 3996

StdP: 12.69

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	4.5	-8.1	4.1	0.6	-3.0	5.4	6.5	11.0	26.0	8.8	28.5	1.6	90	0.453

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.0	89.1	65.0	86.4	63.5	83.9	62.4	67.1	85.3	65.3	82.9	63.7	80.7	3.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	91.3	76.6	58.4	84.8	74.6	56.7	79.5	73.3	34.0	85.2	32.4	83.0	31.1	80.8	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	8.8	6.8	-6.6	94.0	8.9	2.4	-13.0	95.7	-18.2	97.1	-23.1	98.5	-29.6	100.2		
			-7.1	70.2	8.5	3.2	-13.2	72.5	-18.2	74.3	-22.9	76.1	-29.1	78.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	25.6	28.8	41.2	50.5	59.7	67.2	70.1	68.1	59.3	49.7	38.4	28.7	
	DBStd	17.42	9.42	9.75	9.30	8.57	6.22	5.26	4.64	5.02	6.63	8.05	9.30	9.33	
	HDD50	2883	756	594	296	97	5	0	0	0	7	104	362	662	
	HDD65	6258	1221	1013	738	438	184	36	11	27	190	476	797	1127	
	CDD50	2536	0	0	23	110	306	516	624	561	287	94	15	0	
	CDD65	436	0	0	0	2	20	101	170	122	20	1	0	0	
	CDH74	5388	0	0	2	67	329	1187	1861	1489	400	51	2	0	
	CDH80	1642	0	0	0	8	59	355	648	484	80	8	0	0	
Wind	WSAvg	2.6	2.3	2.4	2.9	3.2	3.1	2.9	2.8	2.6	2.6	2.4	2.3	2.2	
Precipitation	PrecAvg	12.9	0.6	0.8	1.5	2.1	2.1	1.3	0.7	0.4	0.5	1.0	1.1	0.9	
	PrecMax	20.9	1.5	1.9	3.0	6.3	4.6	4.9	4.0	1.2	1.7	2.6	3.1	3.5	
	PrecMin	9.0	0.0	0.2	0.7	0.3	0.6	0.2	0.0	0.0	0.0	0.0	0.1	0.1	
	PrecStd	2.6	0.4	0.4	0.6	1.1	1.0	0.9	0.8	0.4	0.4	0.6	0.7	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	56.6	72.6	81.1	85.3	90.8	92.7	91.7	86.1	79.8	70.4	55.1	
		MCWB	41.9	44.2	51.2	59.0	59.8	63.8	68.2	65.9	61.8	56.1	51.5	42.0	
	2%	DB	46.0	50.2	67.2	75.5	81.2	87.4	89.6	88.0	82.1	73.6	62.8	49.3	
		MCWB	36.7	40.6	49.5	54.9	58.9	62.7	66.6	64.5	59.9	53.6	47.0	38.7	
	5%	DB	41.6	45.8	61.7	70.8	77.6	84.3	86.4	85.4	78.9	69.4	56.9	44.9	
		MCWB	34.3	37.5	47.7	53.0	57.9	61.6	64.5	63.5	58.7	51.8	44.7	36.5	
	10%	DB	37.8	41.7	56.4	66.1	74.1	80.9	83.6	82.4	74.9	64.5	51.5	40.8	
		MCWB	32.0	35.3	45.4	51.3	56.7	60.7	63.4	62.2	56.7	49.8	42.3	34.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.9	45.3	53.7	61.4	62.9	67.1	70.7	68.4	64.4	57.9	52.8	43.3
			MCDB	52.6	54.9	68.4	75.2	78.8	84.7	88.1	87.3	81.8	76.3	66.7	52.9
2%		WB	37.4	41.3	50.6	57.5	60.5	64.6	68.1	66.1	61.5	55.0	48.6	39.5	
		MCDB	45.2	49.3	64.3	71.4	76.3	82.1	86.3	84.8	79.3	71.6	60.5	48.0	
5%		WB	34.7	38.3	48.3	54.6	59.1	63.0	66.1	64.3	59.2	52.9	45.5	36.9	
		MCDB	40.8	44.6	60.4	67.3	74.4	80.4	83.9	82.9	76.8	66.8	55.2	43.9	
10%		WB	32.3	35.6	45.7	52.4	57.6	61.6	64.3	62.7	57.3	50.7	42.8	34.6	
		MCDB	37.2	41.4	55.7	64.0	72.1	78.3	80.7	80.4	73.9	62.6	50.8	40.4	
Mean Daily Temperature Range	5% DB	MDBR	19.8	19.0	20.7	21.6	23.5	24.3	25.0	26.4	27.2	25.0	20.6	19.7	
		MCDBR	23.4	23.5	29.2	30.0	28.8	28.7	28.2	29.6	31.8	32.2	28.2	24.9	
		MCWBR	16.3	15.9	16.0	14.8	13.3	12.7	13.4	13.8	15.1	16.1	15.5	16.3	
	5% WB	MCDBR	21.9	21.8	26.7	26.1	26.4	26.0	26.5	28.0	30.1	29.1	25.6	23.0	
		MCWBR	15.5	15.2	15.7	14.8	13.5	13.0	13.9	14.4	15.3	15.8	14.7	15.6	
Clear-Sky Solar Irradiance	taub	0.364	0.423	0.524	0.555	0.526	0.516	0.506	0.462	0.439	0.477	0.435	0.356		
	taud	1.918	1.809	1.696	1.690	1.815	1.865	1.900	2.008	2.007	1.839	1.869	1.969		
	Ebn at Noon	226	232	223	230	242	245	246	253	247	212	198	217		
	Edh at Noon	40	53	68	73	66	63	60	52	49	51	42	35		
All-Sky Solar Radiation	RadAvg	655	935	1203	1529	1935	2155	2190	2067	1648	1023	635	541		
	RadStd	44	82	87	147	151	129	138	117	93	137	84	48		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	+1.21	+1.31	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	-1.37	N/A	N/A	N/A	N/A

Nomenclature: See separate page